CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

EGFR: 13 molecules stopped early

21 stopped trials across 13 molecules: Cetuximab, Erlotinib, Osimertinib, Tarloxotinib and 9 more. 19 of the stops were the trial's own verdict and 2 followed a decision taken elsewhere. Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

7.5%
21 of 280 closed trials (95% CI 5.0%–11.2%)

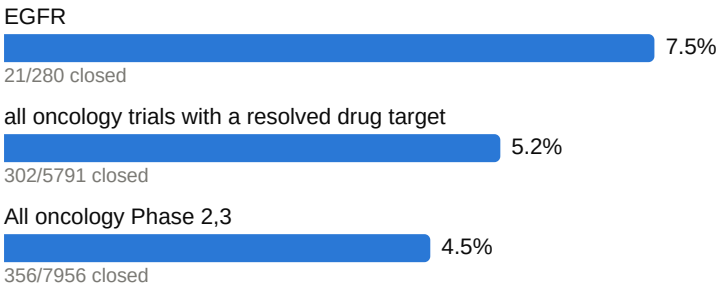
15 / 4 / 0
efficacy / safety / both (adds to 21)

28.9%
of 969 trials have closed · 689 still open or unresolved

3.1%
stopped for a biological reason within 3 years of starting (95% CI 2.0%–4.9%; 366 still at risk; 1.5% area-wide)

4 of 5 **Completed and missed the primary endpoint.** Of 184 completed trials in this segment, 5 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 4 missed and 1 met. Counted separately and never in the rate above; a floor, not a rate. 99 of the 170 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 377 distinct experimental drugs across 969 trials (small molecule 768, antibody 457, unknown 129, adc 61). Most tested: CETUXIMAB (176), OSIMERTINIB (115), PYROTINIB (113), AFATINIB (52), NIMOTUZUMAB (52). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 21 stops came from 20 sponsor-asset programmes across 18 sponsors; the largest (Rain Oncology Inc) contributed 2; removing that programme's trials from both sides leaves 6.9%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 18 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 13.9%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	8	132	6.1%	3.1%–11.5%
2018–2020	6	80	7.5%	3.5%–15.4%
2021–2024	7	68	10.3%	5.1%–19.8%

Where the stops sit

Efficacy / futility: 15, Safety: 4, Benefit–risk: 2.
Sponsors with most stops: Johnson & Johnson (2), Eli Lilly (2), Rain Oncology Inc (2), Genmab (1), Takeda (1), University of Colorado, Denver (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT06685068	2	Genmab GMAB	Anti-EGFR/c-Met Antibody-drug Con…	2026-03	Benefit–risk	Genmab has decided to terminate the GEN1286 program due to an unfavorable benefit-risk profile not supporting further development.
NCT04844073	2	Takeda TAK	TAK-186	2025-06	Efficacy	Sponsor decision based on limited anti-cancer activity of TAK-186.

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05167409	2	University of Colorado, Denver	EVORPACEPT, CETUXIMAB, PEMBROLIZUMAB	2024-10	Safety	Accrual was terminated due to safety concerns
NCT05619172	2	SOTIO Biotech AG	NANRILKEFUSP ALFA, CETUXIMAB	2024-03	Efficacy	Due to lack of efficacy shown at the time of the interim analysis.
NCT04853043	2	University of Utah	CETUXIMAB	2024-02	Efficacy	PI Determination - Slow recruitment, funding concerns, poor response to treatment
NCT05653427	2	Johnson & Johnson JNJ	AMIVANTAMAB	2023-10	Efficacy	N\~15 subjects data analysis did not meet statistical consideration of ORR and study primary/secondary objectives were not met. Enrollment was terminated for futility

The 6 most recent of 21 stopped trials. Every molecule behind all 21 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the diligence report.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 28.9% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta: enter your work email at **clinicaltrialfailures.com/asset-check** and it opens straight away.

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.